

Validity Analysis: LAILA

Target context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Overall risk: **HIGH**

Deployment Context

Use case: Support high school students (G10-12) in editing and revising their Arabic essay drafts before submitting them for grading.

Target population: high school students (G10-12) in Arab countries (especially Qatar)

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	2	Significant gaps	high
Input Content 	2	Significant gaps	high
Input Form 	4	Minor gaps	high
Output Ontology 	1	Serious concern	high
Output Content 	3	Moderate gaps	high
Output Form 	1	Serious concern	high
Average	2.2		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

LAILA is a high-quality Arabic AES dataset that establishes substantial within-Qatar validity for numeric trait scoring on persuasive and explanatory essays, but it is structurally misaligned with the proposed deployment along three of the four dimensions marked HIGH priority. (1) Output Form (1/5) and Output Ontology (1/5) are direct mismatches: the benchmark validates QWK agreement on ordinal scores while the deployment requires open-ended natural-language revision suggestions — an output category absent from the benchmark, and a field-wide gap in Arabic AES. (2) Input Ontology (2/5) covers only 2 of 5–7 deployment-required genres; literary analysis, religious/Quranic commentary, and cultural commentary are entirely absent and the CAST rubric is explicitly persuasive/argumentative. (3) Input Content (2/5) is single-country (Qatar) with explicit author acknowledgement of limited generalizability; cross-national rubric and curriculum alignment with Egypt, Saudi Arabia, Jordan,

and Lebanon is empirically unstudied. Output Content (3/5) is strong within Qatar (qualified annotators, substantial IAA) but unvalidated cross-nationally. Input Form (4/5) aligns well as both contexts are digital Arabic text. The authors themselves prohibit operational deployment without independent validation [Q104, Q106].

ID	Evidence
Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)
Q106	"We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application." (p.10)

Practical Guidance

What This Benchmark Measures

LAILA can validate model fidelity in producing numeric ordinal trait scores (REL, ORG, VOC, STY, DEV, MEC, GRA) and a holistic score on Qatari G10–12 explanatory and persuasive essays, against teacher-annotator judgments anchored in the CAST rubric. It is most informative for Output Content and Input Form within Qatar's persuasive/argumentative scope, where the benchmark's annotation rigor and digital-text alignment with deployment are genuine strengths.

Construct Depth

The benchmark probes scoring agreement at the trait level with substantial QWK (avg 0.66–0.75), but it does not probe whether those scores can be translated into actionable feedback, whether they generalize across Arab national curricula, or whether they extend to genres beyond persuasive/explanatory. Construct depth is therefore moderate within scope and shallow outside it. The narrow REL scale (0–2) further compresses granularity on relevance judgments [Q91].

What Else You Need

For deployment, supplementary evaluations are required for: (a) Output Form — a feedback-generation evaluation harness with rubrics for actionability, MSA register-appropriateness, and comprehensibility, evaluated by Arab teachers and students; (b) Input Ontology — datasets and rubrics covering literary analysis, religious/Quranic commentary, and cultural commentary; (c) Input Content & Output Content — cross-national annotation studies in Egypt, Saudi Arabia, Jordan, and Lebanon to estimate scoring-norm divergence from Qatari CAST conventions; (d) within-Qatar — school-type representativeness audit covering government, Arabic private, international, and community schools.

ID	Evidence
Q91	"Fourth, the relevance trait is scored on a narrow scale from 0 to 2, which limits the granularity of evaluation and may cause QWK to underrepresent subtle differences in model predictions." (p.10)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's task taxonomy is restricted to 8 prompts spanning only two genres — explanatory and persuasive [Q38] — with the CAST rubric explicitly designed for persuasive/argumentative writing [Q118]. The deployment requires coverage of literary analysis, religious/Quranic commentary, and cultural commentary genres common in Arab high-school curricula, which the authors explicitly acknowledge as out of scope: 'limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing' [Q89]. Web research confirms this is a field-wide gap: TAQEEEM 2025 and ZaQQ 2025 maintain the same explanatory/persuasive restriction [WEB-10, WEB-11]. The benchmark also organizes its evaluation around prompt-specific and cross-prompt scoring setups [Q14, Q46], with the entire model output taxonomy restricted to numeric trait scoring [Q48, Q181] — no benchmarked task category corresponds to natural-language revision suggestion generation that the deployment requires. Given IO is HIGH priority in elicitation and multiple genre categories are missing, this represents significant taxonomic underrepresentation.

ID	Evidence
Q14	"We establish baseline AES results for Arabic under two evaluation setups: prompt-specific and cross-prompt." (p.2)
Q38	"LAILA comprises 7,859 essays collected across 8 distinct prompts: 3 explanatory and 5 persuasive (3,446 and 4,413 essays, respectively)." (p.6)
Q46	"For each setup, we investigate two research questions in the context of Arabic AES: (RQ1) how do different categories of models compare in overall performance?, and (RQ2) which models achieve the best results across individual traits?." (p.7)
Q48	"The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly." (p.7)
Q89	"Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing." (p.10)
Q118	"Table 6: CAST Persuasive/Argumentative Writing Rubric - English Translation (Bashendy et al., 2024)." (p.13)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
WEB-10	TAQEEEM 2025 — first Arabic AES shared task — also limited to scoring outputs (https://aclanthology.org/2025.arabicnlp-sharedtasks.134/)
WEB-11	ZaQQ 2025 dataset also confined to standard essay genres; no Arabic AES resource for religious/literary genres (https://www.mdpi.com/2306-5729/10/9/148)

Strengths

- Within its persuasive/argumentative scope, the taxonomy is well-articulated with two complementary evaluation setups (prompt-specific and cross-prompt) that probe different generalization regimes [Q14, Q46]
- The benchmark explicitly recognizes cross-prompt evaluation as more realistic and prioritizes it for future research [Q72, Q77], aligning with deployment-style use

ID	Evidence
Q14	"We establish baseline AES results for Arabic under two evaluation setups: prompt-specific and cross-prompt." (p.2)

Q46	“For each setup, we investigate two research questions in the context of Arabic AES: (RQ1) how do different categories of models compare in overall performance?, and (RQ2) which models achieve the best results across individual traits?.” (p.7)
Q72	“We note that the applicability of the prompt-specific setup is relatively limited due to reliance on target prompt essays that are labeled, which is often unavailable in practice.” (p.9)
Q77	“Overall, the findings of the benchmarking experiments above highlight the need to prioritize future research on cross-prompt AES, which is practically closer to the real world, while best models are still far from human performance.” (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Deployment requires explanatory, persuasive, literary analysis, religious/Quranic commentary, cultural commentary, and possibly narrative/descriptive genres, plus a feedback-generation task category alongside scoring [elicitation A4; regional YAML genre_coverage].

IO-2: Yes — benchmark covers only explanatory and persuasive genres [Q38, Q89]; literary analysis, religious text commentary, and cultural commentary are entirely absent. The output task taxonomy is also limited to numeric scoring [Q48], omitting the natural-language feedback generation category that the deployment requires.

IO-3: No clear evidence that benchmark categories are irrelevant to the regional context — the persuasive and explanatory genres are themselves part of Arab curricula; the issue is omission, not inclusion of foreign categories.

IO-4: Documented gaps: (a) genre coverage missing literary, religious, cultural commentary [Q89]; (b) absence of feedback-generation task type [Q48, Q181]; (c) explanatory prompts P1 and P5 themselves showed interpretive instability with higher REL=0 frequencies [Q124, Q125], suggesting even within-scope coverage has weaknesses.

ID	Evidence
Q38	“LAILA comprises 7,859 essays collected across 8 distinct prompts: 3 explanatory and 5 persuasive (3,446 and 4,413 essays, respectively).” (p.6)
Q48	“The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly.” (p.7)
Q89	“Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing.” (p.10)
Q124	“However, P1 and P5 display relatively higher frequencies at score 0, compared to other prompts, suggesting potential ambiguity in prompt wording that may have limited clear interpretation.” (p.14)
Q125	“Notably, both P1 and P5 are explanatory prompts, which often allow greater flexibility in structure and content.” (p.14)
Q181	“In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics.” (p.18)
WEB-10	TAQEEM 2025 — first Arabic AES shared task — also limited to scoring outputs (https://aclanthology.org/2025.arabicnlp-sharedtasks.134/)
WEB-11	ZaQQ 2025 dataset also confined to standard essay genres; no Arabic AES resource for religious/literary genres (https://www.mdpi.com/2306-5729/10/9/148)

Information Gaps

- Whether the deployment's specific genre mix (e.g., proportion of religious vs. literary essays) can be quantified for each target country's curriculum
- Whether explanatory prompts in LAILA approximate any deployment genre adequately enough to provide partial signal

Requires Expert Verification

- Qatari and other Arab MoE curriculum specialists should confirm the genre breakdown actually assigned at G10–12 across school types
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Remediation

Gap 1: Only explanatory and persuasive prompts are covered; literary analysis, religious/Quranic commentary, and cultural commentary genres are absent — and confirmed missing from TAQEEM 2025 and ZaQQ as well.

Recommendation: Collect a complementary Arabic essay corpus covering the missing genres at G10–12, with prompts validated against multiple national curricula, and develop genre-appropriate annotation guidelines.

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Content was collected from 24 Qatari high schools with 4,372 students under exam-style supervision [Q3, Q27], with prompts deliberately designed to be 'developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics' [Q99] and gender balance pursued [Q23, Q100]. However, the deployment targets Egypt, Saudi Arabia, Jordan, Lebanon and other Arab countries [elicitation A1], and the authors themselves acknowledge that single-country sampling 'may limit its generalizability to other Arabic-speaking populations due to diverse educational systems' [Q88]. The Qatar expatriate mitigation is partial only [Q88]. Web research confirms that secondary-curriculum Arabic writing rubrics in Egypt, Saudi Arabia, Jordan, and Lebanon are unverified against CAST [WEB-5, WEB-6, WEB-7], that Lebanese students show documented MSA-register avoidance [WEB-9], and that within Qatar itself ~60% of secondary students attend private schools across 23 curricula [WEB-1, WEB-3, WEB-4] with no documented school-type breakdown of the LAILA sample. Two prompts (P3, P4) are also under-represented [Q90], and average length is only 171 words [Q92], restricting extended-writing assessment. Given IC is HIGH priority, multiple substantive content gaps weigh heavily.

ID	Evidence
Q3	"Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1)
Q23	"[D3] Achieving Gender Balance: We intentionally aimed to balance the number of essays from male and female students to promote fairness, reduce gender-based bias, support inclusive evaluation, and enhance model generalization." (p.3)
Q27	"Data collection spanned one academic year through 27 school visits (24 initial, three follow-ups), by nine team members, to meet the target student count." (p.5)
Q88	"First, LAILA was collected from students in a single country, Qatar, and specific grade levels, grades 10 to 12, which may limit its generalizability to other Arabic-speaking populations due to diverse educational systems. However, this concern is partially mitigated by Qatar's large population of Arab expatriates, resulting in student participants representing a wide variety of Arabic dialects and backgrounds." (p.9)
Q90	"Third, two of the prompts (Prompts 3 and 4) have fewer essays than the rest, which could introduce imbalance and lead to skewed performance in prompt-specific evaluations." (p.10)
Q92	"Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions." (p.10)
Q99	"Essay prompts were carefully designed to be developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics or potentially harmful content." (p.10)
Q100	"The dataset includes balanced gender representation and covers grades 10-12." (p.10)
WEB-1	Qatar 2024–2025: 278 government schools (~137,048 students), 351 private schools (~228,488 students); ~60% of secondary students in private schools (https://grokipedia.com/page/List_of_schools_in_Qatar)
WEB-3	https://www.tutorchase.com/blog/the-education-system-in-qatar-explained
WEB-4	23 distinct curricula operate in Qatar; British curriculum leads at 45.8% of K-12 schools (https://consulting-haus.com/wp-content/uploads/2023/09/CH_Education_Sector-Overview_September-2023.pdf)
WEB-5	Egypt's curriculum reform suggests evolving writing-assessment norms not aligned with CAST (https://www.impact-se.org/middle-eastern-curriculum-reform-a-window-into-national-values/)
WEB-6	Saudi Arabia uses centrally produced rating tools managed by supervisors (https://www.moe.gov.sa/ar/education/studies/Documents/Education%20in%20Saudi%20Arabia.pdf)
WEB-7	Lebanon's CRDP authoritative but Arabic writing rubrics not in searchable form (https://wenr.wes.org/2017/05/education-in-lebanon)

WEB-9	Lebanese student MSA-register avoidance suggests differing teacher norms (https://www.researchgate.net/publication/267714826_Foreign_Language_Education_in_Lebanon_A_Context_of_Cultural_and_Curricular_Complexities)
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Strengths

- Authentic classroom-produced writing under exam-style supervised conditions with IRB approval [Q21, Q3, Q95]
- Deliberate cultural relevance and age-appropriateness in prompt design [Q99]
- Explicit gender balance principle [Q23, Q100]
- Largest publicly available Arabic AES dataset [Q1, Q79]

ID	Evidence
Q1	"LAILA, the largest publicly available Arabic AES dataset to date, comprising 7,859 essays annotated with holistic and trait-specific scores on seven dimensions: relevance, organization, vocabulary, style, development, mechanics, and grammar." (p.1)
Q3	"Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1)
Q21	"[D1] Ensuring Data Integrity: We designed LAILA to capture authentic, classroom-produced writing created without external assistance. This design choice preserves genuine linguistic variation, learner errors, and authentic complexities essential for real-world AES applications." (p.3)
Q23	"[D3] Achieving Gender Balance: We intentionally aimed to balance the number of essays from male and female students to promote fairness, reduce gender-based bias, support inclusive evaluation, and enhance model generalization." (p.3)
Q79	"To bridge this gap, this paper introduced LAILA, the first large-scale publicly available dataset for Arabic AES." (p.9)
Q95	"The collection of LAILA received approval from an Institutional Review Board (IRB Number: QU-IRB 159/2024-EA) and was conducted with informed consent from both students and their legal guardians." (p.10)
Q99	"Essay prompts were carefully designed to be developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics or potentially harmful content." (p.10)
Q100	"The dataset includes balanced gender representation and covers grades 10-12." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document essay content issues harming content validity

Checklist responses:

IC-1: Yes — Arabic high-school essays inherently require cultural, dialectal, and curricular knowledge. LAILA's content is anchored in Qatari institutional norms (QUTC, Qatar MoE) [Q33, Q84], and the regional YAML notes that Levantine-, Egyptian-, and French-contact-influenced MSA may diverge from Qatar's Gulf MSA conventions [WEB-9, gap_id 4].

IC-2: Partially — Qatar-targeted prompts are culturally appropriate for Qatar [Q99] but unvalidated for Egypt, Saudi Arabia, Jordan, or Lebanon, where curriculum bodies and rubrics differ [WEB-5, WEB-6, WEB-7] and no cross-national rubric comparison has been published (gap_id 1).

IC-3: No clear Western-specific content was identified; LLM selection used the Open Arabic LLM Leaderboard [Q178], and prompts are Arabic. The risk is Qatar-centric rather than Western-centric content.

IC-4: INSUFFICIENT DOCUMENTATION — the LAILA paper does not document recruitment of non-Qatari Arab annotators; the regional YAML explicitly flags this as a gap requiring expert elicitation in each target country (gap_id 1).

IC-5: Documented content issues: single-country sourcing [Q88], unverified cross-national applicability, prompt-count imbalance for P3/P4 [Q90], short average length restricting long-form assessment [Q92], and undocumented Qatari school-type representation [WEB-1, WEB-4].

ID	Evidence
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q84	"We heartily thank our dedicated annotators for their contributions and express our gratitude to Qatar University Testing Center, the Ministry of Education and Higher Education in Qatar (the Arabic Section of the Department Of Educational Supervision in particular), participating schools, and students for making this work possible." (p.9)
Q88	"First, LAILA was collected from students in a single country, Qatar, and specific grade levels, grades 10 to 12, which may limit its generalizability to other Arabic-speaking populations due to diverse educational systems. However, this concern is partially mitigated by Qatar's large population of Arab expatriates, resulting in student participants representing a wide variety of Arabic dialects and backgrounds." (p.9)
Q90	"Third, two of the prompts (Prompts 3 and 4) have fewer essays than the rest, which could introduce imbalance and lead to skewed performance in prompt-specific evaluations." (p.10)
Q92	"Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions." (p.10)
Q99	"Essay prompts were carefully designed to be developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics or potentially harmful content." (p.10)
Q178	"The selection of the LLMs was based on the Open Arabic LLM Leaderboard, where we selected" (p.17)
WEB-1	Qatar 2024-2025: 278 government schools (~137,048 students), 351 private schools (~228,488 students); ~60% of secondary students in private schools (https://grokipedia.com/page/List_of_schools_in_Qatar)
WEB-4	23 distinct curricula operate in Qatar; British curriculum leads at 45.8% of K-12 schools (https://consulting-haus.com/wp-content/uploads/2023/09/CH_Education_Sector-Overview_September-2023.pdf)
WEB-5	Egypt's curriculum reform suggests evolving writing-assessment norms not aligned with CAST (https://www.impact-se.org/middle-eastern-curriculum-reform-a-window-into-national-values/)
WEB-6	Saudi Arabia uses centrally produced rating tools managed by supervisors (https://www.moe.gov.sa/ar/education/studies/Documents/Education%20in%20Saudi%20Arabia.pdf)
WEB-7	Lebanon's CRDP authoritative but Arabic writing rubrics not in searchable form (https://wenr.wes.org/2017/05/education-in-lebanon)
WEB-9	Lebanese student MSA-register avoidance suggests differing teacher norms (https://www.researchgate.net/publication/267714826_Foreign_Language_Education_in_Lebanon_A_Context_of_Cultural_and_Curricular_Complexities)

Information Gaps

- School-type breakdown of LAILA's 24-school sample (government vs. Arabic private vs. international vs. community)
- Whether and how non-Qatari Arab dialectal influence on MSA is treated by annotators
- Cross-national curriculum rubric divergence for STY/DEV

Requires Expert Verification

- Curriculum specialists in Egypt, Saudi Arabia, Jordan, Lebanon to confirm whether Qatari prompts and topic conventions transfer
- LAILA authors regarding within-Qatar school-type sampling

Remediation

Gap 1: All essays are from Qatari students; cross-national generalizability is acknowledged as limited and within-Qatar school-type breakdown is undocumented.

Recommendation: Conduct cross-national essay collection in Egypt, Saudi Arabia, Jordan, and Lebanon at G10-12, and audit the Qatar sample's school-type distribution against MoEHE statistics (government, Arabic private, international, community).

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The input form is digital Arabic text submitted via Microsoft Forms under a 65-minute timed condition [Q25, Q26], aligning well with the deployment's text-based formative-feedback interface. The script is native Arabic (RTL), eliminating Latin-script mismatch concerns. Length distribution is unimodal around 171 words [Q134, Q135] with a smaller cluster of short essays (11–60 words, 1,061 essays) [Q136], and very few essays exceed 500 words [Q138] — a meaningful but minor mismatch with deployment, where formative drafts may vary more in length and grade-level. Model-specific input formats (AraBERT pooling + 816 features [Q156, Q157], AraT5 instruction+text [Q148], LLMs with 4096-token context [Q184]) are reasonable for Arabic NLP. Internet penetration in Qatar is 99% [WEB-14], supporting digital deployment. IF was marked LOWER priority, and the form alignment is genuinely strong; the main residual concern is that benchmark essays are exam-style timed productions while deployment drafts are pre-submission and may differ in length/polish.

ID	Evidence
Q25	"[D5] Ensuring Authentic Writing Conditions: To emulate real academic or high-stakes testing scenarios, LAILA was designed to collect essays written under timed conditions using a digital submission platform. This setup ensures consis" (p.3)
Q26	"taining one persuasive and one explanatory prompt, along with a unique, pre-generated ID to maintain students' anonymity while enabling metadata tracking [D6]. We also configured a Microsoft Form for digital essay submission, set with a 65-minute time limit to enforce a time-restricted setup [D5]." (p.5)
Q134	"The data exhibits a unimodal pattern centered around an average essay length of 171 words, with the majority of essays concentrated between 90 and 210 words." (p.15)
Q135	"A pronounced peak occurs in the 150 to 180 word range, where the count reaches its highest at 1,065 essays." (p.15)
Q136	"Shorter essays are also represented, with 1,061 essays falling between 11 and 60 words." (p.15)
Q138	"Beyond 420 words, the frequency drops sharply, with only a few essays exceeding 500 words, suggesting that longer essays are less common in the dataset." (p.15)
Q148	"The input to the model consisted of an instruction to score the essay along with the essay text, and the model was fine-tuned to generate the trait names followed by their corresponding scores." (p.16)
Q156	"The model is fine-tuned with a custom regression head and combined with handcrafted features for trait scoring." (p.16)
Q157	"The essay and prompt were provided as input to the encoder, with the max-pooled token representation concatenated with the handcrafted features." (p.16)
Q184	"The number of examples was fixed to 5, balancing scoring context with the context length limit (4096 tokens)." (p.18)
WEB-14	99% internet penetration in Qatar as of 2024 (https://www.researchgate.net/figure/Total-Number-of-Schools-in-Qatar-Primary-Intermediate-Secondary_fig1_369912994)

Strengths

- Native Arabic script, RTL-appropriate, no script-encoding mismatch [Q25, Q26]
- Digital text submission matches deployment modality
- Diverse handcrafted feature set covering surface, readability, lexical, semantic, syntactic aspects [Q54]
- Qatar internet penetration ~99% supports web-based deployment [WEB-14]

ID	Evidence
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Q25	"[D5] Ensuring Authentic Writing Conditions: To emulate real academic or high-stakes testing scenarios, LAILA was designed to collect essays written under timed conditions using a digital submission platform. This setup ensures consis" (p.3)
Q26	"taining one persuasive and one explanatory prompt, along with a unique, pre-generated ID to maintain students' anonymity while enabling metadata tracking [D6]. We also configured a Microsoft Form for digital essay submission, set with a 65-minute time limit to enforce a time-restricted setup [D5]." (p.5)
Q54	"We implemented the 816 handcrafted features introduced by Sayed et al. (2025) for Arabic AES, encompassing surface, readability, lexical, semantic, and syntactic aspects." (p.8)
WEB-14	99% internet penetration in Qatar as of 2024 (https://www.researchgate.net/figure/Total-Number-of-Schools-in-Qatar-Primary-Intermediate-Secondary_fig1_369912994)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions match — both benchmark and deployment use digital Arabic text. Length distribution differs slightly: benchmark concentrates at 150–180 words [Q135] with few long essays [Q138], whereas deployment drafts may span a wider range.

IF-2: Yes — digital infrastructure is available; Qatar reports 99% internet penetration [WEB-14]. Secondary-geography figures are not verified [regional YAML internet_connectivity_secondary_geographies].

IF-3: Domain-specific differences: timed exam-style production [Q25] vs. formative untimed draft; benchmark lacks long-form samples (>500 words) [Q138, Q92], which limits validation for extended writing assignments.

IF-4: Minor form mismatches: length-distribution skew toward short essays and absence of long-form coverage [Q92, Q138]; timed vs. untimed production conditions.

ID	Evidence
Q25	"[D5] Ensuring Authentic Writing Conditions: To emulate real academic or high-stakes testing scenarios, LAILA was designed to collect essays written under timed conditions using a digital submission platform. This setup ensures consis" (p.3)
Q92	"Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions." (p.10)
Q134	"The data exhibits a unimodal pattern centered around an average essay length of 171 words, with the majority of essays concentrated between 90 and 210 words." (p.15)
Q135	"A pronounced peak occurs in the 150 to 180 word range, where the count reaches its highest at 1,065 essays." (p.15)
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Information Gaps

- Length and polish distribution of typical formative drafts in deployment
- Internet/device infrastructure in non-Qatar target geographies

Remediation

Gap 1: Benchmark concentrates on short timed exam essays (~171 words) with very few long-form samples; deployment drafts may differ in length and polish.

Recommendation: Augment evaluation with longer-form essay samples (>500 words) and untimed draft conditions to test model robustness for formative pre-submission contexts.

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

This is the most severe misalignment. LAILA's output ontology is exclusively numeric ordinal trait scoring on the CAST rubric: 6 traits at 0–5 plus REL at 0–2 [Q34, Q36], with holistic computed as the sum [Q35, Q182]. All benchmarked models 'followed a multitask setup predicting holistic and trait scores jointly' [Q48]; LLMs produce trait scores in JSON format [Q181]; AraT5 formulates AES as text generation of trait names plus scores [Q147]. The deployment requires actionable natural-language revision suggestions explaining why points were lost [elicitation A3, regional YAML output_function], a qualitatively different output category that is entirely absent from the benchmark's output schema. Web research confirms this gap is field-wide: 'crucial aspects of writing evaluation, such as providing feedback, have often been overlooked' even in English [WEB-12], and no Arabic rubric-to-natural-language feedback system exists [WEB-13, WEB-23]. The CAST rubric is also explicitly genre-scoped to persuasive/argumentative writing [Q118], compounding the misalignment with deployment genres. The narrow REL scale (0–2) further limits granularity [Q91]. OO is HIGH priority and the mismatch is direct, structural, and acknowledged by the authors who 'explicitly prohibit its use for high-stakes educational decisions' [Q104].

ID	Evidence
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q35	"Additionally, a Holistic score (HOL) was computed as the sum of the 7 trait scores." (p.5)
Q36	"Six traits (all but REL) were rated on a 6-point scale (0 = lowest, 5 = highest), while REL was rated on a 3-point scale (0 = not relevant, 1 = partially relevant, and 2 = fully relevant)." (p.5)
Q48	"The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly." (p.7)
Q91	"Fourth, the relevance trait is scored on a narrow scale from 0 to 2, which limits the granularity of evaluation and may cause QWK to underrepresent subtle differences in model predictions." (p.10)
Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)
Q118	"Table 6: CAST Persuasive/Argumentative Writing Rubric - English Translation (Bashendy et al., 2024)." (p.13)
Q147	"The AES task is formulated as a text generation problem, where the model predicts the trait scores sequentially." (p.16)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
Q182	"The holistic score was computed as the sum of the individual trait scores." (p.18)
WEB-12	AES literature explicitly identifies feedback generation as an overlooked area (https://link.springer.com/article/10.1007/s10462-024-11017-5)
WEB-13	AAEE Saudi system provides corrected essays but not trait-grounded revision suggestions (https://www.researchgate.net/publication/n/343796441_Automated_students_arabic_essay_scoring_using_trained_neural_network_by_e-jaya_optimization_to_support_personalized_system_of_instruction)
WEB-23	No Arabic system bridges trait scores to actionable natural-language feedback for secondary students (https://www.sciencedirect.com/science/article/abs/pii/S0306457319304935)

Strengths

- Trait-decomposed scoring offers a principled construct decomposition (REL, ORG, VOC, STY, DEV, MEC, GRA) that could anchor downstream feedback generation [Q34]
- CAST rubric is institutionally grounded in QUTC and provided in Arabic [Q33, Q114]
- Zero-rule for irrelevant essays (REL=0 → all traits 0) reflects pedagogical logic [Q37, Q117]

ID	Evidence
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q37	"Furthermore, if an essay received a REL score of 0, all other trait scores were set to 0, as irrelevant responses are not subject to further evaluation." (p.5)
Q114	"For annotating LAILA, we adopted the rubric from the Core Academic Skills Test (CAST), which is provided in Arabic, developed by the Qatar University Testing Center (QUTC)." (p.12)
Q117	"Note: A score of 0 is assigned if the student does not attempt the task, provides a response that falls below the performance level described for score 1, or submits content that is not relevant to the topic of the given prompt." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The 7-trait label space is broadly relevant to Arab high-school writing assessment, but the rubric is explicitly designed for persuasive/argumentative writing [Q118], limiting regional relevance for literary, religious, or cultural-commentary genres.

OO-2: Missing categories: (a) natural-language feedback / revision-suggestion outputs that the deployment requires [elicitation A3, WEB-12]; (b) trait categories or rubric dimensions specific to genres absent from the benchmark (e.g., textual fidelity for Quranic commentary, literary device recognition).

OO-3: INSUFFICIENT DOCUMENTATION on whether CAST trait definitions encode Qatar-specific stylistic norms versus pan-Arab conventions; the regional YAML flags this as gap_id 1.

OO-4: Stakeholder-driven taxonomy redesign would be needed to (i) add a feedback-generation output category and (ii) extend or replace the rubric for non-persuasive genres.

OO-5: Documented taxonomy issues: numeric-only output schema [Q48, Q181, Q147]; CAST genre-specificity [Q118]; narrow REL scale limits granularity [Q91]; no validated mapping from trait scores to actionable feedback.

ID	Evidence
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q48	"The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly." (p.7)
Q91	"Fourth, the relevance trait is scored on a narrow scale from 0 to 2, which limits the granularity of evaluation and may cause QWK to underrepresent subtle differences in model predictions." (p.10)
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Q147	"The AES task is formulated as a text generation problem, where the model predicts the trait scores sequentially." (p.16)
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WEB-12	AES literature explicitly identifies feedback generation as an overlooked area (https://link.springer.com/article/10.1007/s10462-024-11017-5)

Information Gaps

- Whether trait-score outputs can be reliably converted to feedback through downstream generation models
- Whether CAST trait definitions encode Qatar-specific norms

Requires Expert Verification

- Pedagogical experts on whether the CAST trait decomposition supports formative feedback at all

Remediation

Gap 1: Output schema is restricted to numeric trait scoring with no feedback-generation category; CAST rubric is genre-specific to persuasive/argumentative writing.

Recommendation: Redesign the output ontology to include a feedback-generation task and extend or replace CAST with rubrics validated for literary analysis, religious commentary, and cultural commentary genres. Engage Qatari and other Arab curriculum specialists to define rubric categories.

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation quality within Qatar is a relative strength: 6 annotators + 3 supervisors, all Arabic-language teachers/lecturers with 5 holding advanced degrees [\[Q30, Q31\]](#), with detailed guidelines, training, moderation, and adjudication procedures [\[Q41, Q42, Q43\]](#). Inter-annotator agreement is substantial, averaging QWK 0.66 (P5) to 0.75 (P1) [\[Q44\]](#). However, P5 required third-annotator adjudication on 23.3% of essays [\[Q45\]](#), indicating interpretive instability on explanatory prompts. The decisive concern is institutional homogeneity: rubric, annotators, schools, and funding are all Qatar-anchored (QUTC, QRDI, Qatar MoE) [\[Q33, Q84, Q86\]](#), with no documented non-Qatari annotators and no cross-national norm validation (gap_id 1, gap_id 6). For deployment in Egypt, Saudi Arabia, Jordan, and Lebanon, it is unknown whether STY and DEV labels reflect pan-Arab teacher consensus or Qatar-specific institutional convention [\[WEB-5, WEB-6, WEB-7\]](#). OC was rated MODERATE priority, and the in-region (Qatar) signal is genuinely positive — but the cross-national transferability is unsubstantiated. A score of 3 reflects strong within-Qatar content validity tempered by unvalidated transferability.

ID	Evidence
Q30	"The hired annotation team consisted of 6 annotators and 3 supervisors, all of whom were Arabic language teachers or lecturers with educational backgrounds." (p.5)
Q31	"Five members of the team hold advanced degrees (MSc or PhD) in the Arabic language." (p.5)
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q41	"To ensure consistency, 2 supervisors developed an annotation guidebook with detailed terminology, exemplars, and annotated practices for each prompt type. Annotators were required to review these materials and complete training sessions before formal annotation. Moderation sessions followed, where discrepancies were discussed and interpretations of the rubric were harmonized." (p.6)
Q42	"The scoring process was conducted using the Assessment Gourmet Platform, which supports large-scale, anonymized essay scoring. The platform ensured that annotators were blinded to student identity [D6], while only supervisors had access to annotator metadata for monitoring purposes." (p.6)
Q43	"The essays were randomly distributed among the annotators to minimize systematic bias, and the scoring sessions were capped to prevent annotators' fatigue. Each essay was independently scored by two annotators across all traits. If the difference in HOL scores between the two annotators was less than 6 points, the mean of the two scores was computed and then rounded down to the nearest integer; this rounded value was adopted as the final score for each trait. However, essays with large discrepancies between annotators (≥ 6 points difference in the HOL score) were flagged and escalated to a supervisor (a third annotator in this case), who provided the final adjudicated score and offered feedback to the annotators to improve alignment and consistency in subsequent batches of essays." (p.6)
Q44	"We compute IAA using Quadratic Weighted Kappa (QWK) (Williamson et al., 2012) to assess agreement between the two main annotators (A1 and A2), prior to adjudication. Table 4 reports QWK per trait and prompt, with agreement strength following Landis and Koch (1977). Overall, IAA was substantial across prompts, with an average QWK ranging from 0.66 (P5) to 0.75 (P1), showing strong consistency." (p.6)
Q45	"However, variability emerges by prompt and trait. In particular, P5 showed the lowest average agreement, noting that it has the highest percentage of essays that require a third annotator (A3: 23.3%; Table 3). This suggests greater initial disagreement" (p.6)
Q84	"We heartily thank our dedicated annotators for their contributions and express our gratitude to Qatar University Testing Center, the Ministry of Education and Higher Education in Qatar (the Arabic Section of the Department Of Educational Supervision in particular), participating schools, and students for making this work possible." (p.9)
Q86	"This work was made possible by NPRP grant NPRP14S-0402-210127 from the Qatar Research Development and Innovation (QRDI) Council." (p.9)

WEB-5	Egypt's curriculum reform suggests evolving writing-assessment norms not aligned with CAST (https://www.impact-se.org/middle-eastern-curriculum-reform-a-window-into-national-values/)
WEB-6	Saudi Arabia uses centrally produced rating tools managed by supervisors (https://www.moe.gov.sa/ar/education/studies/Documents/Education%20in%20Saudi%20Arabia.pdf)
WEB-7	Lebanon's CRDP authoritative but Arabic writing rubrics not in searchable form (https://wenr.wes.org/2017/05/education-in-lebanon)

Strengths

- All annotators qualified Arabic-language educators, 5 with MSc/PhD [Q30, Q31]
- Substantial inter-annotator QWK agreement averaging 0.66–0.75 [Q44]
- Comprehensive guidelines with terminology, exemplars, training, moderation [Q41, Q102]
- Annotator-blind scoring with anonymization [Q42, Q98]
- Adjudication procedure for high-discrepancy essays [Q43]
- Fair compensation for annotators at or above local professional rates [Q101]

ID	Evidence
Q30	"The hired annotation team consisted of 6 annotators and 3 supervisors, all of whom were Arabic language teachers or lecturers with educational backgrounds." (p.5)
Q31	"Five members of the team hold advanced degrees (MSc or PhD) in the Arabic language." (p.5)
Q41	"To ensure consistency, 2 supervisors developed an annotation guidebook with detailed terminology, exemplars, and annotated practices for each prompt type. Annotators were required to review these materials and complete training sessions before formal annotation. Moderation sessions followed, where discrepancies were discussed and interpretations of the rubric were harmonized." (p.6)
Q42	"The scoring process was conducted using the Assessment Gourmet Platform, which supports large-scale, anonymized essay scoring. The platform ensured that annotators were blinded to student identity [D6], while only supervisors had access to annotator metadata for monitoring purposes." (p.6)
Q43	"The essays were randomly distributed among the annotators to minimize systematic bias, and the scoring sessions were capped to prevent annotators' fatigue. Each essay was independently scored by two annotators across all traits. If the difference in HOL scores between the two annotators was less than 6 points, the mean of the two scores was computed and then rounded down to the nearest integer; this rounded value was adopted as the final score for each trait. However, essays with large discrepancies between annotators (≥ 6 points difference in the HOL score) were flagged and escalated to a supervisor (a third annotator in this case), who provided the final adjudicated score and offered feedback to the annotators to improve alignment and consistency in subsequent batches of essays." (p.6)
Q44	"We compute IAA using Quadratic Weighted Kappa (QWK) (Williamson et al., 2012) to assess agreement between the two main annotators (A1 and A2), prior to adjudication. Table 4 reports QWK per trait and prompt, with agreement strength following Landis and Koch (1977). Overall, IAA was substantial across prompts, with an average QWK ranging from 0.66 (P5) to 0.75 (P1), showing strong consistency." (p.6)
Q98	"Annotation was conducted on a platform that prevented annotators from accessing student identities or demographic information." (p.10)
Q101	"All annotators were qualified Arabic language educators who received fair compensation at or above local professional rates." (p.10)
Q102	"They completed structured training on annotation guidelines, and inter-annotator disagreements were resolved through systematic adjudication procedures." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators

OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Labels reflect Qatari Arabic-language-teacher perspectives [Q30, Q31, Q33, Q84]. Whether they reflect Egyptian, Saudi, Jordanian, or Lebanese stakeholder perspectives is undocumented [gap_id 1].

OC-2: Potential disagreement is plausible for subjective traits (STY, DEV) but unmeasured. Lebanese student MSA-register avoidance [WEB-9] suggests a different teacher community may judge stylistic features differently.

OC-3: Annotator demographics partially documented: 6+3 team, Arabic-language educators, 5 with advanced degrees [Q30, Q31]; nationality and per-trait IAA breakdown not in published paper [gap_id 6].

OC-4: Re-annotation by a representative cross-national pool would be needed before deploying outside Qatar.

OC-5: Aggregation rule (mean rounded down, supervisor adjudication if HOL diff ≥ 6) [Q43] could erase minority annotator perspectives, though escalation provides a safety valve. Within-Qatar perspectives may also collapse to a single institutional norm.

OC-6: Documented label issues: single-institution annotator pool [Q84], unmeasured cross-national norm transferability, P5 elevated adjudication rate [Q45].

ID	Evidence
Q30	"The hired annotation team consisted of 6 annotators and 3 supervisors, all of whom were Arabic language teachers or lecturers with educational backgrounds." (p.5)
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WEB-5	Egypt's curriculum reform suggests evolving writing-assessment norms not aligned with CAST (https://www.impact-se.org/middle-eastern-curriculum-reform-a-window-into-national-values/)
WEB-6	Saudi Arabia uses centrally produced rating tools managed by supervisors (https://www.moe.gov.sa/ar/education/studies/Documents/Education%20in%20Saudi%20Arabia.pdf)
WEB-7	Lebanon's CRDP authoritative but Arabic writing rubrics not in searchable form (https://wenr.wes.org/2017/05/education-in-lebanon)
WEB-9	Lebanese student MSA-register avoidance suggests differing teacher norms (https://www.researchgate.net/publication/267714826_Foreign_Language_Education_in_Lebanon_A_Context_of_Cultural_and_Curricular_Complexities)

Information Gaps

- Per-trait IAA breakdown (especially STY, DEV)
- Annotator nationality / non-Qatari Arab participation
- Cross-national teacher-judgment correlation studies

Requires Expert Verification

- Sample re-annotation by Egyptian, Saudi, Jordanian, Lebanese teachers to estimate cross-national QWK
-

Remediation

Gap 1: All annotators are Qatar-affiliated; cross-national norm transferability for subjective traits (STY, DEV) is empirically unstudied.

Recommendation: Recruit a representative pool of Egyptian, Saudi, Jordanian, and Lebanese Arabic-language teachers to re-annotate a stratified subsample of LAILA essays; report cross-national QWK per trait to quantify norm divergence before deploying outside Qatar.

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output-form mismatch is the most direct and explicitly documented gap. LAILA evaluates models using QWK between human and model numeric scores [Q58] with hyperparameter optimization on average QWK [Q59, Q60, Q61]; LLMs are constrained to JSON-formatted scores via the outlines library [Q190]. The deployment requires open-ended natural-language revision suggestions in MSA accessible to G10–12 students [elicitation A3, regional YAML output_function]. QWK on numeric scores cannot capture pedagogical usefulness, actionability, or comprehensibility of generated feedback. The authors themselves recognize this limit by prohibiting high-stakes use [Q104] and recommending 'fairness auditing and stakeholder consultation before real-world application' [Q106]. Web research confirms no Arabic rubric-to-natural-language feedback evaluation framework exists [WEB-12, WEB-13, WEB-23]. OF was marked HIGH priority. Even basic infrastructure for evaluating feedback quality (rubrics for actionability, register-appropriateness, MSA fluency) is absent from the benchmark.

ID	Evidence
Q58	"We evaluate model performance using QWK to measure agreement between human and model scores." (p.8)
Q59	"Hyperparameters are tuned via Bayesian optimization with the Tree-structured Parzen Estimator algorithm (Bergstra et al., 2011), implemented using Optuna's TPESampler." (p.8)
Q60	"We ran 20 trials with 5 startup trials and a fixed random seed of 11." (p.8)
Q61	"The best configuration, selected based on average QWK on the development set, was used for final evaluation on the test data." (p.8)
Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)
Q106	"We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application." (p.10)
Q190	"For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format." (p.18)
WEB-12	AES literature explicitly identifies feedback generation as an overlooked area (https://link.springer.com/article/10.1007/s10462-024-11017-5)
WEB-13	AAEE Saudi system provides corrected essays but not trait-grounded revision suggestions (https://www.researchgate.net/publication/343796441_Automated_students_arabic_essay_scoring_using_trained_neural_network_by_e-jaya_optimization_to_support_personalized_system_of_instruction)
WEB-23	No Arabic system bridges trait scores to actionable natural-language feedback for secondary students (https://www.sciencedirect.com/science/article/abs/pii/S0306457319304935)

Strengths

- QWK is the established gold-standard metric for ordinal scoring agreement [Q58]
- JSON output enforcement and three-seed averaging for LLM experiments improve numeric-output reproducibility [Q189, Q190]
- Authors transparently flag the limits of the current evaluation [Q104, Q106]

ID	Evidence
Q58	"We evaluate model performance using QWK to measure agreement between human and model scores." (p.8)
Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)

Q106	"We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application." (p.10)
Q189	"To account for variability in example selection, each experiment was repeated three times with random seeds 1, 11, and 42, and we report the average performance across runs." (p.18)
Q190	"For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format." (p.18)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: No — benchmark output modality is structured numeric/JSON scores [Q58, Q181, Q190]; deployment requires open-ended natural-language MSA feedback. Direct mismatch.

OF-2: Not applicable — deployment is text-based; TTS not required.

OF-3: Literacy is high in target population (G10–12 students); MSA register-appropriateness for adolescents is the key accessibility consideration, on which the benchmark provides no signal.

OF-4: Documented form mismatch: QWK [Q58] cannot capture pedagogical usefulness, comprehensibility, or actionability of generated revision suggestions; benchmark explicitly prohibits operational deployment without independent validation [Q104, Q105].

ID	Evidence
Q58	"We evaluate model performance using QWK to measure agreement between human and model scores." (p.8)
Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)
Q105	"Users must commit to: (1) preventing re-identification attempts, (2) avoiding demographic profiling or inference, and (3) refraining from deployment in operational assessment contexts without independent validation." (p.10)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
Q190	"For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format." (p.18)
WEB-12	AES literature explicitly identifies feedback generation as an overlooked area (https://link.springer.com/article/10.1007/s10462-024-11017-5)
WEB-13	AAEE Saudi system provides corrected essays but not trait-grounded revision suggestions (https://www.researchgate.net/publication/343796441_Automated_students_arabic_essay_scoring_using_trained_neural_network_by_e-jaya_optimization_to_support_personalized_system_of_instruction)

Information Gaps

- Whether any supplementary Arabic feedback-generation evaluation rubric is in development
- Whether downstream LLM-generated feedback can be calibrated against LAILA scores

Requires Expert Verification

- Pedagogical experts and Arabic-language teachers on what 'good' formative feedback looks like in MSA for G10–12

Remediation

Gap 1: Benchmark evaluates only numeric/JSON score agreement (QWK), providing no signal on natural-language feedback quality, actionability, or MSA register-appropriateness.

Recommendation: Construct a supplementary feedback-generation evaluation: pair each LAILA essay with teacher-authored model revision feedback, then evaluate generated feedback on a multi-axis rubric (faithfulness to trait scores, actionability, MSA register, student comprehensibility). Pilot with Arab high-school teachers and G10–12 students before any deployment.